FORMAL UNRAMIFIEDNESS OF A MORPHISM OF ALGEBRAIC SPACES IS DETECTED ON ETALE CHARTS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2723-2743 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2745-2763, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally unramified morphisms of algebraic spaces, etale charts, and etale locality of formal unramifiedness; the descent is retained.

Source reference. Original lemma lemma-composition-formally-smooth-etale-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2486–2495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-etale-on-top: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2567–2576. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-formally-permanence: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2666–2675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-representable-property-formally-property: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2529–2537. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-definition-formally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2763–2763. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

(1) *f is formally unramified,*
(2) *for every diagram*

$$\begin{array}{ccc} U & \xrightarrow{\psi} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes and the vertical arrows are étale the morphism of schemes ψ is formally unramified (as in More on Morphisms, Definition more-morphisms-definition-formally-unramified), and
(3) *for one such diagram with surjective vertical arrows the morphism ψ is formally unramified.*

Proof. Assume f is formally unramified. By Lemma lemma-representable-property-formally-property the morphisms $U \to X$ and $V \to Y$ are formally unramified. Thus by Lemma lemma-composition-formally-smooth-etale-unramified the composition $U \to Y$ is formally unramified. Then it follows from Lemma lemma-formally-permanence that $U \to V$ is formally unramified. Thus (1) implies (2). And (2) implies (3) trivially

Assume given a diagram as in (3). By Lemma lemma-representable-property-formally-property the morphism $V \to Y$ is formally unramified. Thus by Lemma lemma-composition-formally-smooth-etale-unramified the composition $U \to Y$ is formally unramified. Then it follows from Lemma lemma-etale-on-top that $X \to Y$ is formally unramified, i.e., (1) holds. □